

DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

Academic Year: 2025-26

Year: Second Year

Term: I

Roll No.: 1012411011

Name: Riddhee Kulkarni

Subject: java

Assignment No.: 3

Title:

Date: 08/08/2025

Java
Assignment - 3

Name: Riddhee Kulkarni

PRN: 1012411011

Aim:

Implement a class hierarchy using all types of inheritance & demonstrate method overriding in the context of an employee management system.

Theory:

Inheritance is an OOPs feature of Java where one class acquires properties & methods from another.

1. Single inheritance
2. Hierarchical inheritance
3. Multilevel
4. Multiple
5. Hybrid

Flowchart:



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```
import java.util.Scanner;

class Employee {
    int empID;
    double basicSalary;
    String name, department;

    Employee(String name, int empID, double basicSalary, String department) {
        this.name = name;
        this.empID = empID;
        this.basicSalary = basicSalary;
        this.department = department;
    }

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Employee ID: " + empID);
        System.out.println("Department: " + department);
        System.out.println("Basic Salary: " + basicSalary);
    }
}

class Manager extends Employee {
    double HRA;
    double DA;
    double award;

    Manager(String name, int empID, double basicSalary, String department, boolean outstanding) {
        super(name, empID, basicSalary, department);
        this.HRA = 0.20 * basicSalary;
        this.DA = 0.10 * basicSalary;
        this.award = outstanding ? 5000 : 0;
    }

    @Override
    void display() {
        super.display();
        System.out.println("HRA: " + HRA);
        System.out.println("DA: " + DA);
    }
}
```

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```
        System.out.println("Outstanding Award: " + award);
    }
}

class SeniorManager extends Manager {
    int teamSize;

    SeniorManager(String name, int empID, double basicSalary, String department, boolean outstanding, int teamSize) {
        super(name, empID, basicSalary, department, outstanding);
        this.teamSize = teamSize;
    }

    @Override
    void display() {
        super.display();
        System.out.println("Team Size: " + teamSize);
    }
}

class Developer extends Employee {
    String language;
    double HRA, DA, award;

    Developer(String name, int empID, double basicSalary, String department, String language, boolean outstanding) {
        super(name, empID, basicSalary, department);
        this.language = language;
        this.HRA = 0.15 * basicSalary;
        this.DA = 0.08 * basicSalary;
        this.award = outstanding ? 3000 : 0;
    }

    @Override
    void display() {
        super.display();
        System.out.println("Language: " + language);
        System.out.println("HRA: " + HRA);
        System.out.println("DA: " + DA);
        System.out.println("Outstanding Award: " + award);
    }
}
```

```
class Tester extends Employee {
    String tool;
    double HRA, DA, award;

    Tester(String name, int empID, double basicSalary, String department, String tool, boolean outstanding) {
        super(name, empID, basicSalary, department);
        this.tool = tool;
        this.HRA = 0.12 * basicSalary;
        this.DA = 0.09 * basicSalary;
        this.award = outstanding ? 2000 : 0;
    }

    @Override
    void display() {
        super.display();
        System.out.println("Testing Tool: " + tool);
        System.out.println("HRA: " + HRA);
        System.out.println("DA: " + DA);
        System.out.println("Outstanding Award: " + award);
    }
}

class AdminPanel {
    double HRF, PF, PT;

    void SalaryBreakdown(Employee emp, double HRA, double DA, double award) {
        double grossSalary = emp.basicSalary + HRA + DA + award;
        this.PF = 0.05 * emp.basicSalary;
        this.PT = 0.02 * emp.basicSalary;
        this.HRF = 1000;
        double netSalary = grossSalary - (PF + PT + HRF);
        System.out.println("~ Salary Breakdown ~");
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("PF: " + PF);
        System.out.println("PT: " + PT);
        System.out.println("HRF: " + HRF);
        System.out.println("Net Salary: " + netSalary);
    }
}
```

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```
public class EmployeeManagementSystem {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        AdminPanel admin = new AdminPanel();

        System.out.println("Enter Manager Details");
        System.out.print("Name: ");
        String mName = sc.nextLine();
        System.out.print("Employee ID: ");
        int mID = sc.nextInt();
        System.out.print("Basic Salary: ");
        double mSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String mDepartment = sc.nextLine();
        System.out.print("Is Performance Outstanding? (true/false): ");
        boolean mOutstanding = sc.nextBoolean();
        Manager m = new Manager(mName, mID, mSalary, mDepartment, mOutstanding);

        System.out.println("\nEnter Senior Manager Details");
        sc.nextLine();
        System.out.print("Name: ");
        String smName = sc.nextLine();
        System.out.print("Employee ID: ");
        int smID = sc.nextInt();
        System.out.print("Basic Salary: ");
        double smSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String smDepartment = sc.nextLine();
        System.out.print("Is Performance Outstanding? (true/false): ");
        boolean smOutstanding = sc.nextBoolean();
        System.out.print("Team Size: ");
        int teamSize = sc.nextInt();
        SeniorManager sm = new SeniorManager(smName, smID, smSalary, smDepartment, smOutstanding, teamSize);

        System.out.println("\nEnter Developer Details");
        sc.nextLine();
```

```
        System.out.print("Name: ");
        String dName = sc.nextLine();
        System.out.print("Employee ID: ");
        int dID = sc.nextInt();
        System.out.print("Basic Salary: ");
        double dSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String dDepartment = sc.nextLine();
        System.out.print("Programming Language: ");
        String language = sc.nextLine();
        System.out.print("Is Performance Outstanding? (true/false): ");
        boolean dOutstanding = sc.nextBoolean();
        Developer dev = new Developer(dName, dID, dSalary, dDepartment, language, dOutstanding);

        System.out.println("\nEnter Tester Details");
        sc.nextLine();
        System.out.print("Name: ");
        String tName = sc.nextLine();
        System.out.print("Employee ID: ");
        int tID = sc.nextInt();
        System.out.print("Basic Salary: ");
        double tSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String tDepartment = sc.nextLine();
        System.out.print("Testing Tool: ");
        String tool = sc.nextLine();
        System.out.print("Is Performance Outstanding? (true/false): ");
        boolean tOutstanding = sc.nextBoolean();
        Tester tester = new Tester(tName, tID, tSalary, tDepartment, tool, tOutstanding);

        System.out.println("\n--- Employee Details ---");

        System.out.println("\nManager:");
        m.display();
        admin.SalaryBreakdown(m, m.HRA, m.DA, m.award);
```

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```
System.out.println("\nSenior Manager:");
sm.display();
admin.SalaryBreakdown(sm, sm.HRA, sm.DA, sm.award);

System.out.println("\nDeveloper:");
dev.display();
admin.SalaryBreakdown(dev, dev.HRA, dev.DA, dev.award);

System.out.println("\nTester:");
tester.display();
admin.SalaryBreakdown(tester, tester.HRA, tester.DA, tester.award);

sc.close();
}
}
```

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```
Enter Manager Details
Name: Riddhee
Employee ID: 1
Basic Salary: 150000
Department: Marketing
Is Performance Outstanding? (true/false): true

Enter Senior Manager Details
Name: NAndini
Employee ID: 2
Basic Salary: 170000
Department: Product
Is Performance Outstanding? (true/false): true
Team Size: 5

Enter Developer Details
Name: Neeraj
Employee ID: 3
Basic Salary: 130000
Department: Product
Programming Language: java
Is Performance Outstanding? (true/false): true

Enter Tester Details
Name: Shardul
Employee ID: 4
Basic Salary: 130000
Department: Product
Testing Tool: python
Is Performance Outstanding? (true/false): true

--- Employee Details ---

Manager:
Name: Riddhee
Employee ID: 1
Department: Marketing
Basic Salary: 150000.0
HRA: 30000.0
DA: 15000.0
Outstanding Award: 5000.0
~ Salary Breakdown ~
Gross Salary: 200000.0
PF: 7500.0
PT: 3000.0
HRE: 1000.0
Net Salary: 188500.0
```

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```
Senior Manager:
Name: Nandini
Employee ID: 2
Department: Product
Basic Salary: 170000.0
HRA: 34000.0
DA: 17000.0
Outstanding Award: 5000.0
Team Size: 5
~ Salary Breakdown ~
Gross Salary: 226000.0
PF: 8500.0
PT: 3400.0
HRE: 1000.0
Net Salary: 213100.0

Developer:
Name: Neeraj
Employee ID: 3
Department: Product
Basic Salary: 130000.0
Language: java
HRA: 19500.0
DA: 10400.0
Outstanding Award: 3000.0
~ Salary Breakdown ~
Gross Salary: 162900.0
PF: 6500.0
PT: 2600.0
HRE: 1000.0
Net Salary: 152800.0

Tester:
Name: Shardul
Employee ID: 4
Department: Product
Basic Salary: 130000.0
Testing Tool: python
HRA: 15600.0
DA: 11700.0
Outstanding Award: 2000.0
~ Salary Breakdown ~
Gross Salary: 159300.0
PF: 6500.0
PT: 2600.0
HRE: 1000.0
Net Salary: 149200.0
```

Conclusion:

This program demonstrates how to use inheritance & method overriding.